

● Edexcel A-Level Maths · Paper 1: Pure Mathematics 1 · 9MA0/01

Pure 1 — Full Worked Solutions

Wednesday 3 June 2026 · Advanced GCE Level 3

Sat the paper and want to know how you did? Here are **StudyHours model solutions** to all 15 questions — every method step shown the way an examiner wants it, so you can mark your own script and see exactly where the marks were. Built by tutors, not a calculator.

All 15 questions, full method

Answers boxed in green. Common slips flagged in red. "Show all working" questions worked without calculator shortcuts.

1 Transformations of $y = f(x)$, point $P(-4, 5)$

4 marks

(a) $y = 4f(2x)$

$2x$ inside $\rightarrow$ horizontal stretch scale factor $\frac{1}{2}$: x -coord halved. $\times 4$ outside $\rightarrow$ vertical stretch scale factor 4: y -coord $\times 4$.

$x: -4 \rightarrow -2$ $y: 5 \rightarrow 20$

$(-2, 20)$

(b) $y = f(x + 3) - 2$

$x+3 \rightarrow$ translate left 3: $-4 \rightarrow -7$. $-2 \rightarrow$ translate down 2: $5 \rightarrow 3$.

$(-7, 3)$

2 Iteration · $f(x) = e^{2x-1} + 3x - 7$

5 marks

(a) Show α lies in $[1.1, 1.2]$

$$f(1.1) = e^{1.2} + 3.3 - 7 = 3.320 - 3.7 = -0.380 < 0$$

$$f(1.2) = e^{1.4} + 3.6 - 7 = 4.055 - 3.4 = +0.655 > 0$$

Sign change and f is continuous $\rightarrow$ a root α lies in $[1.1, 1.2]$. ■

(b) $x_{n+1} = \frac{1}{2}(1 + \ln(7 - 3x_n))$, $x_1 = 1.1$

$$x_2 = \frac{1}{2}(1 + \ln(7 - 3.3)) = \frac{1}{2}(1 + \ln 3.7)$$

$x_2 = 1.1542$ (4 d.p.)

Continue iterating ($x_3 = 1.1318$, $x_4 = 1.1411$, ... converging):

$\alpha = 1.1384$ (4 d.p.)

3 Differentiation & tangent · $y = x^3 - 2x^2 + 5x - 7$

5 marks

(a) Find dy/dx

$$dy/dx = 3x^2 - 4x + 5$$

(b) Tangent at $x = 2$

y-coord: $8 - 8 + 10 - 7 = 3 \rightarrow$ point $(2, 3)$.

Gradient: $3(4) - 4(2) + 5 = 12 - 8 + 5 = 9$.

$y - 3 = 9(x - 2)$

$$y = 9x - 15 \quad (m = 9, c = -15)$$

4 Functions · $f(x) = 3x^2 - 2$, $g(x) = (5x - 1)/(x - 4)$

5 marks

(a) Range of f

Min at $x = 0$ gives -2 ; parabola opens up. $f(x) \geq -2$

(b) $gf(2)$

$f(2) = 12 - 2 = 10$, then $g(10) = (50 - 1)/(10 - 4) = 49/6$

$$gf(2) = 49/6$$

(c) $g^{-1}(x)$

Let $y = (5x-1)/(x-4) \rightarrow y(x-4) = 5x-1 \rightarrow yx - 5x = 4y - 1 \rightarrow x = (4y-1)/(y-5)$

$$g^{-1}(x) = (4x - 1)/(x - 5)$$

5 Parametric model · $x = 100\sqrt{t}$, $y = 10 \sin(t^2/4)$

5 marks

(a) Maximum height

Max of $10 \sin(\dots)$ is when $\sin = 1$. **10 m**

(b) Horizontal distance when $y = 5$ the second time

$$10 \sin(t^2/4) = 5 \rightarrow \sin(t^2/4) = 1/2$$

Solutions of $t^2/4$: $\pi/6$ (1st time), $5\pi/6$ (2nd time).

$$t^2 = 4 \cdot (5\pi/6) = 10\pi/3 \rightarrow t = \sqrt{10\pi/3}$$

$$x = 100\sqrt{t} = 100 \cdot (10\pi/3)^{1/4}$$

$$x \approx 179.9 \text{ m}$$

SLIP Second time is on the way down: $t^2/4 = 5\pi/6$, not the next full cycle. Easy to take the wrong root.

6 Integration · $\int 3/(7x + 1) dx$

5 marks

(a)

$$(3/7) \ln|7x + 1| + C$$

(b) Find p where $\int_2^p = 6/7$

$$(3/7)[\ln(7p+1) - \ln 15] = 6/7$$

$$\ln((7p+1)/15) = 2 \rightarrow (7p+1)/15 = e^2$$

$$p = (15e^2 - 1)/7$$

7 Segments & Newton–Raphson · semicircle, $\angle BOC = \theta$

7 marks

(a) Show $p \sin \theta + q\theta - 2\pi = 0$

$$\angle AOB = \pi - \theta. \text{ Segment area} = \frac{1}{2}r^2(\text{angle} - \sin \text{angle}).$$

$$R_1 = \frac{1}{2}r^2(\theta - \sin \theta), \quad R_2 = \frac{1}{2}r^2((\pi - \theta) - \sin(\pi - \theta)) \quad [\sin(\pi - \theta) = \sin \theta]$$

$$R_1 = 2R_2: \theta - \sin \theta = 2(\pi - \theta - \sin \theta)$$

$$\theta - \sin \theta = 2\pi - 2\theta - 2\sin \theta \rightarrow 3\theta + \sin \theta - 2\pi = 0$$

$$p = 1, q = 3$$

(b) Newton–Raphson, $\theta_0 = 1.3$

$$f(\theta) = \sin \theta + 3\theta - 2\pi, \quad f'(\theta) = \cos \theta + 3$$

$$f(1.3) = -1.4196, \quad f'(1.3) = 3.2675$$

$$\theta_1 = 1.3 - (-1.4196 / 3.2675)$$

$$\theta_1 = 1.734 \text{ (3 d.p.)}$$

8 Integration from first principles · $x e^{2x}$

5 marks

(a) Express the limit as an integral

$$\int_0^3 x e^{2x} dx$$

(b) Show it equals $Ae^6 + B$

$$\text{By parts (u = x, dv = e}^{2x}\text{dx): } \frac{1}{2}x e^{2x} - \frac{1}{4} e^{2x}$$

$$\text{At 3: } (3/2 - 1/4)e^6 = (5/4)e^6. \quad \text{At 0: } -1/4.$$

$$(5/4)e^6 + 1/4 \rightarrow A = 5/4, B = 1/4$$

9 Trig identity & equation

8 marks

(a) Show $(\cos\theta + \sin\theta)(\operatorname{cosec}\theta - \sec\theta) \equiv k \cot 2\theta$

$$= (\cos\theta + \sin\theta) \cdot (\cos\theta - \sin\theta) / (\sin\theta \cos\theta) = (\cos^2\theta - \sin^2\theta) / (\sin\theta \cos\theta)$$

$$= \cos 2\theta / (1/2 \sin 2\theta) = 2 \cot 2\theta$$

$$k = 2$$

(b) Solve $5(\cos x + \sin x)(\operatorname{cosec} x - \sec x) = 4 \operatorname{cosec}^2 2x$, $-90^\circ < x < 90^\circ$

$$10 \cot 2x = 4(1 + \cot^2 2x) \rightarrow 2 \cot^2 2x - 5 \cot 2x + 2 = 0$$

$$(2 \cot 2x - 1)(\cot 2x - 2) = 0 \rightarrow \cot 2x = 1/2 \text{ or } 2$$

$$\tan 2x = 2 \rightarrow 2x = 63.43^\circ, -116.57^\circ$$

$$\tan 2x = 1/2 \rightarrow 2x = 26.57^\circ, -153.43^\circ$$

$$x = -76.7^\circ, -58.3^\circ, 13.3^\circ, 31.7^\circ \text{ (1 d.p.)}$$

10 Normal & area · $y = 7x / \sqrt{3x^2 + 4}$, P at $x = 2$

11 marks

(a) Equation of normal l

$$\text{Differentiate: } dy/dx = 28(3x^2 + 4)^{-3/2}$$

$$\text{At } x = 2: 16^{3/2} = 64 \rightarrow dy/dx = 28/64 = 7/16. \quad P = (2, 7/2).$$

$$\text{Normal gradient} = -16/7: \quad y - 7/2 = -(16/7)(x - 2)$$

$$32x + 14y - 113 = 0$$

(b) y-coordinate of Q (l meets y-axis)

$$x = 0 \rightarrow 14y = 113$$

$$y = 113/14$$

(c) Exact area of R

$$\text{Area under line: } \int_0^2 (113 - 32x)/14 \, dx = 162/14 = 81/7$$

$$\text{Area under curve: } \int_0^2 7x/\sqrt{3x^2+4} \, dx = (7/3)[\sqrt{16} - \sqrt{4}] = 14/3$$

$$R = 81/7 - 14/3 = (243 - 98)/21$$

$$R = 145/21$$

11 R cos(x+α) & electricity model

11 marks

(a) $4\cos x - 13\sin x = R \cos(x + \alpha)$

$$R = \sqrt{4^2 + 13^2} = \sqrt{185}, \quad \tan \alpha = 13/4$$

$$R = \sqrt{185}, \quad \alpha = 1.273 \text{ rad (3 d.p.)}$$

(b) Minimum demand & time

$$D = 30 + \sqrt{185} \cos(\pi t/12 + 0.2 + \alpha); \text{ min when } \cos = -1.$$

$$D_{\min} = 30 - \sqrt{185} \approx 16.4 \text{ GW}$$

$$\text{Occurs when } \pi t/12 + 1.479 = \pi \rightarrow t \approx 6.35 \approx 06:21$$

(c) Rate k at 3 pm (t = 15)

$$dD/dt = (\pi/12)[-4 \sin(\pi t/12 + 0.2) - 13 \cos(\pi t/12 + 0.2)]$$

$$\text{At } t = 15: k \approx 2.76 \text{ GW/hr (3 s.f.)}$$

(d) Why not the same model every day?

Demand changes with the **seasons** (temperature, daylight hours), so fixed constants can't model every day of the year.

12 Modulus graph & intersections · $y = |px - q|$, $y = k/x$

7 marks

(a) Sketch of $y = |px - q|$

V-shape, vertex on x-axis at $(q/p, 0)$, crossing the y-axis at $(0, q)$.

(b) Range of k for 3 distinct intersections

$y = k/x$ ($k > 0$) only meets the V for $x > 0$. Right branch always gives 1 point.

Left branch: $q - px = k/x \rightarrow px^2 - qx + k = 0$ needs 2 distinct positive roots $\rightarrow$ discriminant $q^2 - 4pk > 0$.

$$0 < k < q^2/(4p)$$

13 Geometric sequence · 3rd, 4th, 5th terms = $\sin\theta$, $\sqrt{2}\cos\theta$, $\sqrt{3}\cot\theta$

7 marks

(a) Common ratio

Equal ratios: $\sqrt{2}\cos\theta / \sin\theta = \sqrt{3}\cot\theta / (\sqrt{2}\cos\theta) \rightarrow \cos\theta = \sqrt{3}/2 \rightarrow \theta = \pi/6$

$$r = \sqrt{2}\cos\theta / \sin\theta = \sqrt{2} \cdot (\sqrt{3}/2) / (1/2)$$

$$r = \sqrt{6}$$

(b) Sum of first 3 terms

3rd term = $\sin(\pi/6) = 1/2$, so term1 = $1/2/r^2 = 1/12$, term2 = $1/2/r = \sqrt{6}/12$.

$$1/12 + \sqrt{6}/12 + 6/12$$

$$(7 + \sqrt{6})/12$$

14 Differential equation · $dx/dt = x(A - t)$

11 marks

(a) Particular solution

Separate: $\int dx/x = \int (A - t)dt \rightarrow \ln x = At - t^2/2 + C$

$$x = 0.3 \text{ at } t = 0 \rightarrow C = \ln 0.3. \quad x = 0.3 e^{At - t^2/2}$$

$$x = 21 \text{ at } t = 4 \rightarrow e^{4A-8} = 70 \rightarrow A = 2 + \frac{1}{4}\ln 70 (\approx 3.062)$$

$$x = 0.3 e^{At - t^2/2}, \quad A = 2 + \frac{1}{4}\ln 70$$

(b) Maximum concentration

$$\text{Max when } dx/dt = 0 \rightarrow t = A. \quad x_{\max} = 0.3 e^{A^2/2}$$

$$\approx 32.6 \text{ mg L}^{-1} \text{ (3 s.f.)}$$

(c) Value of T when $x = 0.1$

$$e^{AT - T^2/2} = 1/3 \rightarrow T^2 - 2AT - 2\ln 3 = 0 \rightarrow T = A + \sqrt{A^2 + 2\ln 3}$$

$$T \approx 6.46 \text{ (3 s.f.)}$$

15 Proof by contradiction · $a^2 + b^2 = c^2$

4 marks

Complete the proof that a and b cannot both be odd

Assume $a = 2m+1$, $b = 2n+1$. Then

$$a^2 + b^2 = 4(m^2 + m + n^2 + n) + 2 = c^2$$

So c^2 is even $\rightarrow c$ is even $\rightarrow c = 2p \rightarrow c^2 = 4p^2$ (divisible by 4).

But $c^2 = 4(\dots) + 2 \equiv 2 \pmod{4}$, **not** divisible by 4.

Contradiction $\rightarrow a$ and b cannot both be odd. ■

Reading your mark scheme

- **"Show that" questions** give marks for the **method line**, not just the answer — full working on Q7(a), Q8(b), Q9(a), Q15 is where the marks live.
- **Accuracy:** watch rounding instructions — 3 d.p. vs 3 s.f. vs exact form. Losing the final A-mark to rounding is the most common drop.
- **"Solutions relying on calculator technology are not acceptable"** means you must show algebra/calculus — a typed answer scores zero even if correct.

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